

Change2

Cohort 3 Group 5

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Change Report

Change Report [23 marks]:

a) Briefly summarise the processes, tools and conventions your team used to plan, make, keep track of, and review changes to the Assessment 1 deliverables, documentation and code you inherited from the other team. (3 marks, ≤ 1 page)

Our main method of keeping track of changes made is by viewing and tracking activities in our GitHub repository. This way, any changes that are made can be easily tracked, with information about what was changed, why it was changed and who changed it. Those implementing the code had a system to document changes made to the maze game and include justifications. After each change, the developer comments and records the following: The change made, the overall goal and the related deliverable. The comments, in addition to being helpful for tracking changes, were also useful for identifying which Assessment 1 deliverables needed to be updated. Version History on Google Docs allowed us to easily view which deliverables had been updated and what the updates were, allowing us to see if there were any more recent changes made to the code that needed to be documented in the deliverables.

During development, our team used many tools to enable smooth project progression and workflow. These tools included GitHub for code development and documentation of changes made, a shared Google Drive folder for any deliverables, collaborative ideas or notes from meetings, and a WhatsApp group chat for quick progress updates, scheduling and thoughts.

On commencement of Assessment 2, we planned out our ideas for events and what changes needed to be made to the game in order to meet the product brief, for example, the player movement wasn't as smooth as we would have liked and required continuous arrow key presses rather than just a single key press. These meeting notes and changes were documented in a shared Google Doc. We created a shared Google Doc checklist of what needed to be done, used throughout the project, with items being crossed off on completion. This also allowed us to track the project's progress.

According to the other team's implementation document, the two main features not implemented were Game Completion, no user interface was implemented to indicate to the user that they had completed the game, and they had also not implemented the 5-minute timer as required by the product brief. This meant we had those features to implement, plus the leaderboard, achievements and 5 events.

b) For each of the following items, include a brief explanation and justification of any changes made to the other team's Assessment 1 deliverables (pointing to specific sections of the deliverables that have been added/updated). Include the precise URLs of the original (i.e. the other team's) and the updated versions of the deliverables. The updated deliverables do not need to adhere to the page limits for Assessment 1. If there is no change to report, please state and justify why no change was necessary.

(Maximum 5 marks per item, to a maximum total of 20 marks; ≤ 8 pages in total with ≤ 3 pages per item)

- i. Requirements
- ii. Architecture
- iii. Method selection and planning
- iv. Risk assessment and mitigation

FOR CHANGES TO ASSESSMENT 1 DELIVERABLES

Changes Made to Implementation

The initial code provided by the original group did not meet the requirements set out by the initial brief, meaning that a lot of code was missing and therefore had to be extended to meet them. The primary components missing from the original code included no win-condition, missing interactables and that the game would last forever (exceeding 5 minutes).

Comments on GitHub:

- Fix original subcommit
- Create AchievementID

ACHIEVEMENTS

Change Made: Defines Achievement

Overall Goal : Achievement feature

Related deliverable : product brief

Change made: Added the abstract Achievement class containing the fields id, title and description and has the logic for unlocking the achievement.

Overall goal: Achievement feature

Related deliverable: product brief

Change made: Code moved to a different file

Overall goal: Achievement feature

Related deliverable: Product brief

Change made: Creates a class with a map where the achievements are stored and creates a subclass for each achievement so that it can be called anywhere in the game.

Overall goal: Achievement feature

Related deliverable: Product brief

Changes made: java.util.HashMap and java.util.Map have been imported.

Overall goal: Achievement feature

Related deliverable: Product brief

Change made: package added to beginning of code
Overall goal: Achievement feature
Related deliverable: Product brief

Change made: package renamed
Overall goal: Achievement feature
Related deliverable: Product brief

Changes made: Spelling fixed
Overall goal: Achievement feature
Related deliverable: Product brief

Change made: typo in file name fixed
Overall goal: Achievement feature
Related deliverable: Product brief

Change made: Achievement class updated
Overall Goal: Achievement feature
Related deliverable: Product brief

Change made: Achievement manager updated
Overall goal: Achievement feature
Related deliverable: Product brief

Change made: code added to create Achievement manager in the game startup
Overall goal: Achievement feature
Related deliverable: Product brief

Change made: File for achievement pop up made
Overall goal: Achievement feature
Related deliverable: Product brief

Change Made: Changing movement to allow for continued movement while holding down a button, plus diagonal movement
Overall Goal: Base Movement of Player
Related Deliverable: Product Brief

Change Made:
- Corrected an error in folder naming. (Items folder was imported as 'items' in many classes.)
- Set up the headless mode for testing and created relevant tests.
Overall Goal: Testing Framework
Related Deliverable: Test2

Change made: Door to win maze has been added
Overall goal: Exit/win feature
Related deliverable: Product brief

Change made: Exit class that draws temporary exit graphics

Overall goal: Exit/win game feature

Related deliverable: Product brief

Changes made: Code class made which randomly generates a number as one of the numbers for the check in code.

Overall goal: Check in code collected around the map

Related deliverable: Product brief

Change made: CodeManager created which changes the state of collect to true.

Overall goal: Check in code

Related deliverable: Product brief

Change made: 5 codes are made

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Add collectable item for code 1

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Add collectable item for code 2

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Add collectable item for code 3

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Rename Code1Collectable to Code3Collectable

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Add collectable item for code 4

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Add collectable item for code 5

Overall goal: Check in code

Related deliverable: Product brief

Changes made: Add each new item to the itemID file

Overall goal: Check in code

Related deliverable: Product brief

Changes made : made into java file

Overall goal: Check in code

Related deliverable: Product brief

Changes made:

- Implemented code & door: collect 5 digits to allow for player to win.
- Interaction tracker: counts the number of interactions the player had with a specific type.
- Notification system for game progress.
- Ensured everything had appropriate documentation.

Overall goal: Game Completability

Related deliverable: Implementation

Changes made:

- Created new interactable, an energy drink that has the inverse effect of Glue.
- Implemented timer, if time runs out, game is lost.

Overall goal: Game Completability & Interactable Objects

Related deliverable: Implementation

Changes made:

- Two new events: a quiz game and professor assignment hand in, in order to obtain two digits of the code.

Overall goal: Game Completability & Interactable Objects

Related deliverable: Implementation

Changes made:

Created a Duck NPC that will briefly chase the character and slow them down if caught.

Overall goal: Interactions

Related deliverable: Implementation

Changes Made to Requirements

The requirements table produced by the original group had to be updated to include new requirements issued by the customer/updated brief. These requirements were provided with new IDs which could be referred to in other related documents (such as implementation).

The new requirements provided were a scoreboard, achievements and additional events (3 hidden and 5 standard).

The provided requirements were given the IDs:

- UR_SCOREBOARD: A leaderboard with the name and score of the top 5 scores.
- UR_ACHIEVEMENTS: Achievements are implemented when specific events occur.

The reason for these adjustments were to categorise the new user requirements issued by the customer/updated brief.

.Changes Made to Risk

There were no need to update risk documentation, as the risk outlined by the original group covered all angles related to risk and mitigation were covered or were not pertinent to the current state of the project.

Changes Made to Planning

Planning was extended to adjust for the extended time to work on the project, however the tools utilised were different. The original team utilised project management tools such as Trello and brainstorming apps such as Miro to share ideas, whereas the current team used the only GitHub and Google Drive, discussing ideas informally through WhatsApp or in person to brainstorm and plan.

The development cycle only remained largely informal with no specific model.

Changes made to the Architecture

The architecture/project structure of the project provided by the initial team was followed by the current team, as the primary code relating to classes, mechanics and interactions was already laid out. This meant that there was no reason to alter the current system architecture as objects and their related managers were already established.